For teachers only. Do not give this part to students.

Distribution and evidence preservation

1. Teachers mark paper objective answers using the same key and weights. Automatic scoring in the shared guide refers to VL Excel; handwritten answers are not read automatically.
2. Use the same questions, concepts, hints and available time for both groups. The control group uses static images and tables instead of interactive simulations; no extra physical experiment is added.
3. Give out A first. Save the original prediction, reason and design before giving out B. Photograph or copy A; let students keep a copy of their original prediction.
4. In B, complete the four basic sets and three conclusions before the heating page and its conclusion.
5. Keep the first confirmed observations readable. Use one line to cross out a revised answer, or write it in the revision area; do not erase it.
6. Save or collect A and B before giving out C. Let students refer to the saved original prediction and write their learning reflection.
7. D and the complete pack are for teachers. Print each part separately using its page range and give it out at the correct stage, rather than giving the whole pack to students at once.
8. Images use oil layers, shade and texture. Before printing in black and white, check that cloudy and clear mixtures and the two oil-layer thicknesses remain distinguishable.
9. Paper tasks do not generate clicks or system time logs. Score the answers and design evidence; do not convert completion, time or clicks into skill scores.
10. Do not deduct marks from a reasonable, testable original prediction just because its result was wrong. Teachers judge written answers and reflection rather than counting keywords.

Keep weekly worksheets separate from the formal 10-item SPSAI and 4-item CKT pre- and post-tests.

Draft activity scoring guide for researcher review; it is not a validated SPSAI or CKT.

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Objective answers and design reference

Independent variable	Combination of digestive juices
Dependent variable	Appearance of the mixture
Controlled variables	Total volume of oil and water / Reaction temperature / Reaction time / Total volume of liquid added / Method of shaking
03 • Conditions for a fair test	Observe all setups after the same reaction time. Use the same starting amounts of oil and water and the same total volume of added liquid in each setup. Use the same method of shaking for every setup.
1. What does digestive juice X do?	a cloudy emulsion (乳狀液)
2. How does digestive juice X help digestive juice Y act?	increase
3. What can this experiment tell us?	Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol (甘油) and fatty acids (脂肪酸) have formed.
Extension conclusion: the digestive juice that loses its function after boiling is_____.	Digestive juice Y

Control group: add 2 mL of water	Separate oil and water layers (no obvious change)
Add 2 mL of digestive juice X	become cloudy
Add 2 mL of digestive juice Y	Separate oil and water layers (thinner oil layer)
Add 1 mL each of digestive juices X and Y	become clear
Add 1 mL each of digestive juices X and Y (boil X, then cool)	become clear
Add 1 mL each of digestive juices X and Y (boil Y, then cool)	become cloudy

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Reference for written designs

04 • Control group

Use oil–water + 2 mL of water as the control group. Compare it with oil–water + 2 mL of X, + 2 mL of Y, and + 1 mL each of X and Y. Keep the initial amounts of oil and water, total added volume, reaction temperature, reaction time and shaking method the same.

05 • Experiment setup design

Clearly label the four basic setups: control, X, Y and X + Y. For heat treatment, compare combinations in which X or Y is boiled and then cooled to 37°C. Change one factor at a time.

An oil layer, a cloudy mixture or a clear mixture only gives clues about appearance. This alone cannot prove that glycerol (甘油) and fatty acids (脂肪酸) have formed; other chemical tests are needed. This simulation uses set teaching results and does not measure reaction speed.

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Scoring guide

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Item	Maximum	Scoring	Full credit	Partial credit	Zero / missing evidence
Draft activity scoring guide: researcher review needed	32	Research note	Six SPS areas, 4 each; four new-knowledge items, 2 each. Total 32.	Review question difficulty, opportunities and scorer agreement across VLs.	Separate from validated SPSAI / CKT; give paper students the same questions and static data.
Initial observation	2	Teacher	2: accurately describes visible oil/water layers and details, separating observation from guesses.	1: a specific but incomplete description.	0: no useful observation; keep missing older evidence pending rather than inventing it.
Six appearances	2	Answer key	All six answers confirmed; matching appearances / 6 × 2, rounded to two decimal places.	Credit in proportion to the matching appearances.	0 if all answered but none match; a missing set stays unavailable, rather than treating completion as skill.
Independent variable	1	Answer key	Select only the combination of digestive juices.	No partial credit.	0 for an incorrect answer; do not replace missing evidence with 0.
Dependent variable	1	Answer key	Select only the appearance of the mixture.	No partial credit.	0 for an incorrect answer; do not replace missing evidence with 0.

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Scoring guide

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Item	Maximum	Scoring	Full credit	Partial credit	Zero / missing evidence
Controlled variables	2	Answer key	Select all five fixed conditions, with no wrong choices.	Correct controlled choices / 5 × 2; a choice from the wrong category gives 0.	Older versions had three choices; do not infer skill from missing new choices. Review cross-version comparisons.
Original testable prediction and reason	2	Teacher	2: conditions can be compared, prediction can be observed, and reason is reasonable.	1: testable but the reason or conditions are incomplete.	0: not testable or no useful content. A reasonable wrong prediction is not penalised. Missing original evidence stays pending.
Conditions for a fair test	1	Answer key	Select time, initial oil/water and total added volume, and shaking method; do not select more total liquid.	No partial credit.	0 for an incorrect set; a version without this question stays unavailable.
Control group	1	Teacher	1: explains its comparison purpose, proposes a setup without digestive juice, and keeps conditions the same.	Enter only 0 or 1.	0: no useful comparison explanation or an unreasonable reason; missing question stays pending.
Setup quality	2	Teacher	2: clear drawing/photo/text, four labelled sets and reasonable volumes.	1: workable but labels or volumes are incomplete.	0: no workable design; do not penalise a written design.

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Scoring guide

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Item	Maximum	Scoring	Full credit	Partial credit	Zero / missing evidence
Fixed conditions and experiment arrangement	2	Teacher	2: fair comparison based on design/evidence; boiled liquids are cooled; only one factor changes each time.	1: generally reasonable but fixed conditions or heating arrangements are incomplete.	0: comparisons are not valid. Set completion is evidence only; time or clicks do not directly earn marks.
Conclusions, comparisons and limits	4	Answer key	Current three conclusions and the heating conclusion: 1 each, total 4. Older six-question weights remain 0.5, 0.5, 0.5, 0.75, 0.75, 1.	Award each question its stated answer-key credit.	Use the questions belonging to that version; do not demand deleted questions or invent missing older answers.
Data and claims	4	Teacher	4: organises data clearly, compares at least two sets, links a claim and explains limits.	3: clear comparison but weaker limits; 2: useful comparison but weak links; 1: one relevant description.	0: no useful communication. Use reasons, control explanation, labels, table and reflection. Judge clarity without scoring knowledge twice.
Bile (膽汁), emulsification (乳化作用) and surface area	2	Teacher: post-learning reflection only	2: bile (膽汁) has no digestive enzymes; emulsification (乳化作用) is physical digestion (物理消化), and small droplets increase total surface area.	1: partly correct but incomplete explanation.	0: concept unused or wrong; an unsubmitted reflection does not mean zero.
Lipase (脂肪酶), chemical digestion (化學消化) and products	2	Teacher: post-learning reflection only	2: lipase (脂肪酶) speeds up chemical digestion (化學消化), forming glycerol (甘油) and fatty acids (脂肪酸).	1: enzyme action or products partly correct.	0: unused or incorrect; do not score by keywords.

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Scoring guide

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Item	Maximum	Scoring	Full credit	Partial credit	Zero / missing evidence
High temperature and enzyme denaturation	2	Teacher: post-learning reflection only	2: links the boiled-Y observations with protein denaturation and loss of function.	1: knows heat has an effect but does not explain how.	0: concept unused or wrong.
Revising the original explanation using data	2	Teacher: post-learning reflection only	2: addresses whether the original prediction is supported, makes specific comparisons and uses concepts to improve the explanation, with appearance limits.	1: comparison or revision is present but links are incomplete.	0: no useful revision or link to data.
SPS, new knowledge and overall totals	32	Totals	Six SPS areas = 24; four knowledge items = 8. Show overall only after reflection and all required scores are complete.	Blank means pending; 0 means zero. Teacher fields accept only whole numbers in range.	Invalid or non-whole numbers do not produce a valid total. Do not fill blanks with 0 just to complete scoring.
Colours, records and keeping teacher scores	0	Note	Excel uses green for full/correct, red for zero/incorrect, amber for partial; fills show categories.	Original, first and final answers and events are separate; digital time means visible activity time.	Teacher scores remain in the separately saved Excel file, and a new export does not include them. Student PDFs contain no scores.

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